

Date 13-08-2021

Time allowed 1 hr

ANALYTICAL CHEMISTRY
CHEM-306
Regular (B-Group)

Note: Attempt two questions.

Question No1 is compulsory

Q.No1: Answer the following questions (40 marks)

- ✓ Define stoichiometry
- ✓ Derive relationship between mole and no of equivalents
- ✓ Identify the type of reaction: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- ✓ A hydrated compound has an analysis of 76.9% CaSO_4 and 23.1% H_2O . What is the formula?
- ✓ 10M of H_2SO_4 = ? N
- ✓ What amount in grams of carbon dioxide is produced by combustion of 10 gram Glucose?
- ✓ What is the conjugate acid of each of the following Brønsted-Lowry bases?
(a) HCO_3^- (b) CO_3^{2-} (c) OH^- (d) H_2PO_4^-

8 ✓ If 20 gram of hydrochloric acid mixed with 10 gram of sodium hydroxide, which one will be limiting reagent?

Q.No2: a) Define Brønsted acids and bases. Give an example of a conjugate pair in an acid-base reaction.
b) Calculate the pH of a 0.15 M solution of acetic acid (CH_3COOH , $K_a = 1.5 \times 10^{-5}$)

Q.No3: a) Which of the following solutions can be classified as buffer systems?

(1) $\text{KH}_2\text{PO}_4/\text{H}_3\text{PO}_4$, (2) $\text{NaClO}_4/\text{HClO}_4$. Explain your answer.

b) Find the pH and pOH of the following a) 0.1M HCl b) 0.1 NaOH

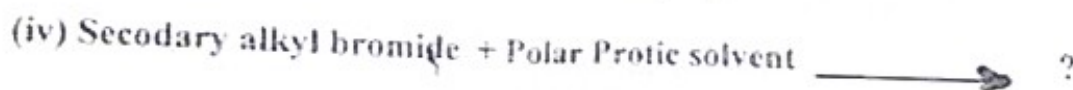
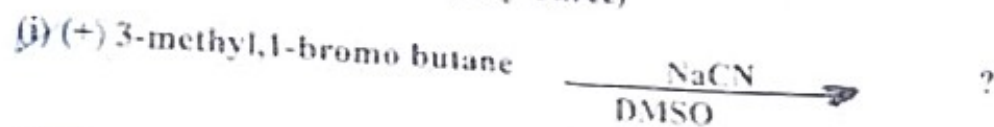
2
521
56

NOTE: Attempt all questions:

Question 1: Justified the following statements: (Any Four)

- (i) The base induced elimination of hydrogen chloride occurs more rapidly from cis than trans 1-chloro-4-methyl cyclohexane.
- (ii) The greater stability of axially substituted conformational isomer were obtained in 2-bromo cyclohexanone or 2,4- di -t-butyl cyclohexanone.
- (iii) Primary alkyl halides with polar aprotic solvents always favor S_N2 substitution reaction.
- (iv) The stereochemistry of cyclohexane has significant importance in all cycloalkane compounds.
- (v) The di equatorial and diaxial conformations of 1,2- dimethyl decalin occur more in cis isomer about 99: 1 at room temperature than trans form.

Question 2: Draw the following reactions with proper mechanism and also mention the name of each reaction? (Any Three)



Question 3: Write short notes on following: (Any Two)

- (i) Racemization.
- (ii) Elimination reactions.
- (iii) Stereochemistry of cyclohexane

To determine power factor of an inductor circuit

Apparatus Chock, Lamp, electric board, auto transformer, and multimeter.

Theory: Transformer is a pair of wire coils in close proximity. The transformer has two coils primary and secondary. Transformer is used to step up or step down AC voltages. Transformer work only with alternating current therefore, there are two types of transformer step-up transformer and step-down transformer.

Power in AC circuit: The power is the product of current and voltage. When the power is positive, the capacitor absorbs energy from the source that drives the current. When the power is negative, the capacitor returns energy to the driving source.

General expression for power in AC circuit.

$$\langle P \rangle = I_{rms} V_{rms} \cos \phi$$

The factor $\cos \phi$ is called the power factor. A purely resistive circuit has a power factor of one, While a circuit with only inductance and capacitance has a power factor of zero. In RLC circuit, the power factor depends on frequency.

$V \rightarrow$ applied voltage.

$V_1 \rightarrow$ Voltage across chock

$V_2 \rightarrow$ Voltage across tanip.

PROCEDURE: Connect the apparatus as shown in figure. Note the readings of V , V_1 , V_2 . In same way increase AC potential and take another reading repeat this procedure at least three times and note the readings of V , V_1 and V_2 each time. Put all calculated values in given formula. In last take mean value of all calculated values of power factor.

DR. M. A. KAZI INSTITUTE OF CHEMISTRY, UNIVERSITY OF SINDH, JAMSHORO

Physical Chemistry (Theory)
BS. P-III & M.Sc (Prev)
(Regular / Group A & B)

1st Semester, Examination 2016

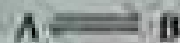
CHEM: 504 / 70
Time Allowed: 2 h
Max. Marks: 60

Attempt any two questions:

(Section - A)

Q. 01. (a) Define Max-well distribution law and discuss velocity distribution curve at different temperatures?
(b) Calculate the temperature in °C at which $\sqrt{2}$ velocity of O_2 gas becomes similar to that of Helium gas molecules at 0 °C.

Q. 02. Derive rate expression for the reaction



OR

Derive rate expression for the reaction



Q. 03. (a) Explain in detail Collision theory of reaction rates.

(b) Calculate activation energy of a reaction whose reaction rate at 27 °C gets doubled for 10 °C rise.

Attempt any two questions:

(Section - B)

Q. No. 1. (a) Explain Latent Heat of vaporization. Derive Clapeyron and Clausius-Clapeyron equation.
(b) Explain in detail the second law of thermodynamics with respect to entropy?

Q. No. 2. (a) What is spectroscopy? Explain the atomic spectrum of hydrogen and hydrogen like atoms.
(b) Explain de-Broglie's work. Derive the de-Broglie's equation and explain its applications.

Q. No. 3. (a) State and Explain Quantum chemistry and its applications?

(b) Derive Schrodinger's wave equation for 1-dimension and 3-dimensions.

Attempt All Questions

Section I (Group-A)

- Q.No.1 State Maxwell's law of distribution of molecular speeds amongst the molecule of a gas?
- Q.No.2 Can the molecules of a gas have high and almost zero speeds? Show that the most probable speed is given by $C_{mp} = \sqrt{2RT/M}$
- Q.No.3 How size of molecule effects on ideal behavior of gas? Drive an expression for excluded volume?
- Q.No.4 Calculate the pressure exerted by 10^{23} gas molecules each with mass 10^{-22} kg in a container of volume 1 dm^3 . The root mean square speed 10^3 ms^{-1} . What is total kinetic energy of the gas molecule? Calculate temperature of gas?
- Q.No.5 If probability "p" is plotted against speed "c" for several temperature, then What will be the most probable distribution velocity, Curve, C_{mp} of diagram of speed of molecules # temperature T_1, T_2, T_3 , where $T_3 > T_2 > T_1$.
- Q.No.6 What are state functions? What are their characteristics? Which of following are state functions?
 (a) Q, (b) W, (c) $E = PV$ (d) Q/W

Q.No.7. Explain following statements

- (a) A gas show cooling effect after adiabatic expansion
 (b) Q and W are not state functions but became state function under certain conditions
 (c) Isothermal reversible work is greater than that for a real gas between the same initial and final state

Q.No.8. Show that dP and dV are exact differentials from the relations a) $PV = RT$, b) $P(V, T)$ and drive cyclic rule in each case

Q.No.9. Prove that $W_{rev} - W_{irr} > 0$

Q.No.10. One mole of an ideal gas at 300K and 30 atm is expanded to double its volume isothermally. Calculate the work of expansion if (a) the expansion were carried out reversibly and (b) irreversibly against a constant opposing pressure of 1.5 atm.

Section II

Q.No.1 Drive rate expression for the reaction $A + B \rightleftharpoons C$

Q.No.2. a) Explain Neil Bohr's atomic model? Drive De-Broglie's equation?

SECTION-B: (BRIEF QUESTIONS)

Note: Attempt all the questions from this section.

Q1. Name the following complexes (5 marks)

- i. $[\text{Co}(\text{NH}_3)_4\text{CO}_2]\text{Cl}$ ii. $[\text{Pt}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]\text{Br}$ iii. $[\text{Cr}(\text{NCS})_4(\text{NH}_3)_2]$ iv. $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$
v. $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$ vi. $\text{K}_3[\text{Ir}(\text{C}_2\text{O}_4)_3]$ vii. $[\text{NH}_3)_4\text{Co}-(\text{NH}_2)_2\text{Co}(\text{NH}_3)_4]^{4+}$
viii. $[\text{NH}_3)_4\text{Co}-\text{NH}_2-\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})]\text{Cl}_5$ ix. $[\text{C}_2\text{O}_4)_2\text{Cr}-(\text{OH})_2-\text{Cr}(\text{C}_2\text{O}_4)_2]^{4+}$ x. $[\text{Cr}(\text{en})_3][\text{Ni}(\text{CN})_5]$

Q2. Write the formula/structure of the following: (5 marks)

- i. tetraamminesulphatocobalt(III) nitrate ii. sodium tetrachlorozincate(II) iii. sodium amminebromochloronitroplatinate(II) iv. dichlorobis(ethylenediamine)nitritocobalt(III) chloride v. octacarbonyl- μ -dicarbonyldiron(0)
vi. bis(acetylacetonato)oxovanadium(IV) vii. sodium oxotetrafluorochromate(IV)
viii. potassium tetrafluoroargentate(I) ix. tetrabromo- μ -dibromodibromodipalladium(II) ion x. hexaaquairon(II) sulphate

Q3. What is CFSE? Calculate the CFSE in terms of Δ_0 for octahedral complexes for d^4 to d^7 electron configuration in case of high-spin and low-spin complexes. (2.5 marks)

Q4. Draw the crystal field splitting energy diagram of octahedral, tetragonal and square planar complexes. (2.5 marks)

Q5. Describe the Platinum and Cobalt complexes by Werner's in the tabulated format.

Hint: Old formula, Werner's Formula, Dissociation, number of particles, and number of electrical charge. (2.5 marks)

Q6. On the basis of VBT, predict the oxidation state of the central metal atom/ion, hybridization, geometry and magnetic behavior of the following complexes. (2.5 marks)

- a) $[\text{Zn}(\text{CN})_4]^{2-}$ b) $[\text{Ni}(\text{NH}_3)_4]^{2+}$ c) $[\text{CoCl}_4]^{2-}$ d) $[\text{Pt}(\text{Cl})_4]^{2-}$ e) $[\text{Fe}(\text{CN})_6]^{3-}$

Q7. Predict the outer-orbital/inner-orbital octahedral complexes by VBT. (2.5 marks)

- a) $[\text{Fe}(\text{NH}_3)_6]^{2+}$ b) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ c) $[\text{MnF}_6]^{3-}$ d) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ e) $[\text{Fe}(\text{CN})_6]^{4-}$

Q8. Write the oxidation state of the central metal in each of the following. (2.5 marks)

- a) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]^+$ b) $\text{K}[\text{CuBr}_2]$ c) $[\text{Co}(\text{py})_5\text{SO}_4\text{Cl}]$ d) $[\text{Cr}(\text{ox})_3]^{3-}$ e) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2\text{H}_2\text{O}$

Q9. What is the EAN? Calculate EAN for the central metal atom in the following complexes and also compare the stability of the complexes. (2.5 marks)

- a) $[\text{Co}(\text{NH}_3)_6]^{2+}$ b) $\text{Cu}(\text{CN})_4]^{3-}$ c) $[\text{Fe}(\text{CN})_6]^{4-}$ d) $[\text{Fe}(\text{CN})_6]^{3-}$ e) $[\text{Ag}(\text{NH}_3)_4]^+$

Q10. Define are flexidentate and ambidentate ligands give an example with structures? Write denticity of the ligands in the following complexes. (3.5 marks)

- a) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ b) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ c) $[\text{Ni}(\text{dmg})_2]$ d) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$ e) $\text{Na}_2[\text{Ni}(\text{edta})]$

Q11. Arrange the following in the increasing order of molar conductivity by Werner's. (01 mark)

- a) $\text{Na}[\text{Cr}(\text{NH}_3)_2\text{Cl}_4]^{2-}$ b) $\text{Cr}(\text{NH}_3)_3\text{Cl}_3$ c) $\text{Na}_2[\text{CrCl}_6]$

Q12. A complex of the composition $\text{PtCl}_2(\text{NH}_3)_3$ has a molar conductance of 404 siemens. Based on this data, guess its proper formula. (01 mark)

Q13. The substance $\text{CoBr}_3 \cdot 4\text{NH}_3 \cdot 2\text{H}_2\text{O}$ behaves like a $(3+1-)$ electrolyte in solution; write its correct formula. (01 mark)

Q14. Explain why Cu^{2+} does not form a regular octahedral complex whereas Ni^{2+} does. (01 mark)

Q15. Since NH_3 is in the middle of the spectrochemical series, some of its complexes are high-spin while others are low-spin. Illustrate. (01 mark)

Q16. Calculate the CFSE in terms of Δ_0 for a d^8 system in octahedral and tetrahedral complexes. Which is more stable? (01 mark)

Q17. Draw the splitting diagram of d^8 electrons of Cu^{2+} ion in tetragonally elongated and compressed distorted octahedral complexes. Also calculate the Jahn-Teller stabilization energy. (02 marks)

Q18. Draw the diagram for the splitting of d-orbitals of d^8 ion in its low-spin and high-spin tetragonally elongated distorted octahedral complexes. Which complex show strong distortion? (01 mark)

Dr. M. A. Kazi Institute of Chemistry, University of Sindh, Jamshoro

BS-III/M.Sc (Prev)

27/4/19

Inorganic Chemistry
Marks: 50

Time: 2hrs

Note: Attempt paper of 50 marks.

Q1. Solve the following reasons (12 Marks compulsory question).

- Cu^{2+} forms distorted octahedral complexes.
- There are no low spin and high spin tetrahedral complexes.
- Cu^{2+} salts are blue and Cu^{+} salts are white.
- $\Delta t = 0.4\Delta_o$
- NH_3 readily forms complex but NH_4^+ does not.
- CFT does not consider any orbital overlap.
- Splitting pattern for d orbitals is $5d > 4d > 3d$.
- $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ ion is more stable than $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$.
- $t_{2g}(\text{unsym}) - e_g(\text{unsym})$ pair does not exist.
- CrO_4^{2-} is d^0 complex and yet it is deeply colored.
- Cr^{3+} forms $[\text{CrCl}_6]^{3-}$ but corresponding complexes for Mn^{3+} and Fe^{3+} are not known.
- $[\text{Ni}(\text{CO})_4]$ is tetrahedral while $[\text{Ni}(\text{CN})_4]^{2-}$ is square planar.

Q2. Discuss the color, no of unpaired electrons and magnetic properties for Fe^{2+} and Mn^{2+} ions.

(10 Marks)

Q3. Calculate CFSE for $[\text{CoF}_6]^{3-}$, $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Cu}(\text{NH}_3)_6]^{2+}$.

(8 Marks)

Q4. Differentiate between Atomic and Molecular orbitals as well as Bonding and antibonding orbitals.

(10 Marks)

Q5. Draw MO diagram for O_2^{2-} and HCl as well as give their BO and EC.

(10 Marks)

Q6. Give important demerits of CFT.

(8 Marks)

THE END

Dr. M.A. KAZI, INSTITUTE OF CHEMISTRY, UNIVERSITY OF SINDH, JAMSHORO
Department of Organic Chemistry
Organic Chemistry (BS-II)-Group-A (Regular) and & Evening

Total Marks 70

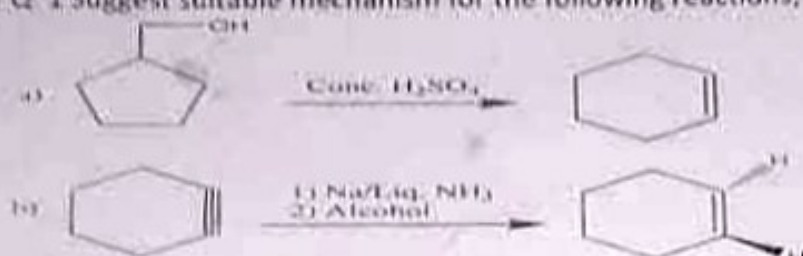
Time allowed 2.0 hr. Dated = 17.12.2019

CHEM-412

Note: Attempt all Questions.

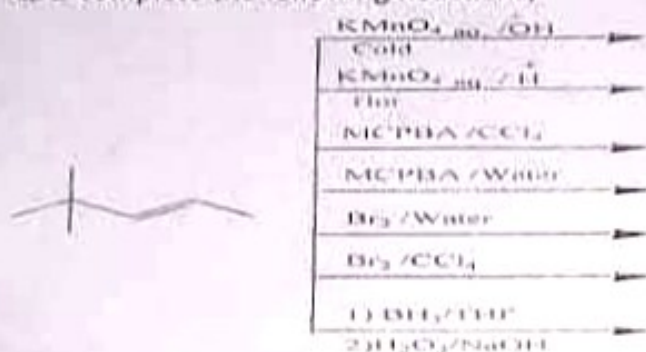
Q. 1 Suggest suitable mechanism for the following reactions;

(10x2 = 20)



Q. 2 Complete the following Reactions;

(7X2=14)



Q. 2 Perform the following conversions;

(5X2 = 10)

- Benzene into n-Propylbenzene
- Propene into Propanal

Q. 3 Suggest suitable reasons to the following facts;

(3x2 = 10)

- Pyrrole is less basic than pyridine
- Ethanol is less acidic than phenol

Q. 4a Draw the hybridized structure of urea

(5)

Q. 4b Identify the type of hybridization of hetero atom in each of the following molecules;

(5)

Aniline, Pyrrole, Pyridine, imine and Enamine

Q. 5 Define the following terms with suitable example

(3X2=6)

- Conjugation
- Hyperconjugation

Best of luck

DR. M. A. KAZI INSTITUTE OF CHEMISTRY
DEPARTMENT OF ANALYTICAL CHEMISTRY
Mid-Term P-III, First Semester

Total Marks = 30

NOTE: Attempt All

Dated: 05-04-2019
Time allowed: 45 min.

- Q.1 What are different types of gravimetric methods? Give at least one example of each. 07
- Q.2 Silver can be determined by forming precipitate of AgCl , why a large excess of Cl^- increases the precipitate's solubility? 03
- Q.3 Differentiate between inclusion and occlusion impurities, how they can be avoided? 05
- Q.4 24.60 mg of $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$ was heated from room temperature to 1000°C at a rate of 5°C min. , thermogram showed a first mass loss of 3.03 g, determine the identity of volatilization product and solid residue. 05
- Q.5 A 1.4639-g sample of limestone was analyzed for Ca by isolating Calcium as CaSO_4 , yielding a precipitate of 1.4058 g. Report the amount of Ca, as %w/w. 05
- Q.6 For what pH range will the following precipitates have their lowest solubility? 05
- a. CaC_2O_4 b. PbCrO_4 c. BaSO_4 d. $\text{Ca}_3(\text{PO}_4)_2$

Attempt All Questions

Section I (Group-A)

- Q.No.1 State Maxwell's law of distribution of molecular speeds amongst the molecule of a gas?
- Q.No.2 Can the molecules of a gas have high and almost zero speeds? Show that the most probable speed is given by $C_{mp} = \sqrt{2RT/M}$
- Q.No.3 How size of molecule effects on ideal behavior of gas? Drive an expression for excluded volume?
- Q.No.4 Calculate the pressure exerted by 10^{23} gas molecules each with mass 10^{-25} kg in a container of volume 1 dm^3 . The root mean square speed 10^3 ms^{-1} . What is total kinetic energy of the gas molecule? Calculate temperature of gas?
- Q.No.5 If probability "p" is plotted against speed "c" for several temperature, then What will be the most probable distribution velocity, C_{ave} , C_{rms} of diagram of speed of molecules at temperature T_1, T_2, T_3 , where $T_3 > T_2 > T_1$.
- Q.No.6 What are state functions? What are their characteristics? Which of following are state functions?
(a) Q, (b) W, (c) $E+PV$ (d) $Q-W$
- Q.No.7 Explain following statements
(a) A gas show cooling effect after adiabatic expansion
(b) Q and W are not state functions but became state function under certain conditions
(c) Isothermal reversible work is greater than that for a real gas between the same initial and final state
- Q.No.8 Show that dP and dV are exact differentials from the relations a) $PV=RT$, b) $P(V-b)=RT$, and drive cyclic rule in each case
- Q.No.9 Prove that, $W_{rev} - W_{ir} > 0$
- Q.No.10 One mole of an ideal gas at 300K and 30 atm is expanded to double its volume isothermally. Calculate the work of expansion if (a) the expansion were carried out reversibly and (b) irreversibly against a constant opposing pressure of 1.5 atm.

Section II

- Q No.1 Drive rate expression for the reaction $A + B \rightleftharpoons C$
- Q No. 2, a) Explain Neil Bohr's atomic model? Drive De-Broglie's equation?

INORGANIC CHEMISTRY, 1st SEMESTER, MID-TERM EXAMINATION PAPER 2022
BS(P-III) GROUP-B, INORGANIC CHEMISTRY THEORY, REGULAR

SUBJECT: INORGANIC CHEMISTRY
 DATE: 21-03-2022
 TIME: 1HR

COURSE # CHEM-500
 MAX. MARKS: 30

Attempt all Questions

Q1 (a). Name of the following complexes? (05 marks)

- i) $[(\text{NH}_3)_5\text{Cr}-\text{OH}-\text{Cr}(\text{NH}_3)_5]\text{Cl}_2$ ii) $[(\text{CO})_5\text{Fe}(\text{CO})_2\text{Fe}(\text{CO})_5]$ iii) $[\text{Cr}(\text{EDTA})]$ iv) $[\text{Ni}(\text{CO})_4]\text{py}$
 v) $(\text{NH}_4)_3[\text{ZrF}_7]$ vi) $\text{Na}[\text{B}(\text{NO}_3)_4]$ vii) $[\text{Cr}(\text{NCS})_4(\text{NH}_3)_2]$ viii) $[(\text{NH}_4)_4\text{Co}(\text{NH}_2)(\text{OH})\text{Co}(\text{NH}_3)_4]^+$
 ix) $[\text{AuCl}(\text{dien})]^{2-}$ x) $[\text{Ni}(\text{NH}_3)_6]_3[\text{Co}(\text{NO}_2)_6]_2$

Q1 (b). Write the structure of the following: (05 marks)

- i) diamminetetraakis(isothiocyanato)chromate(III) ii) tris(ethylenediamine)rhodium(III)ion
 iii) sulphatopentamminecobalt(III) chloride iv) triamminebromochloronitroplatinum(IV) chloride v)
 pentaquonitrosyliron(I) vi) sodiumpentacyanonitrosylferrate(II) vii) tetrammineplatinum(II)
 tetrachloroplatinate(II) viii) ammoniumhexaisothiocyanatochromate(II)
 ix) potassiumtetrafluoroargentate(I) x) lithiumtetrahydroaluminate(III)

Q2. What is denticity of a ligand? Write the denticity, formulas, charge on the ligands with name of donor atoms? (7.5 marks)

- i) diethylenetriamine ii) dimethylglyoximate iii) acetylacetonato iv) glycinate v) Nitrosyl
 vi) thiosulphato vii) Nitrito viii) amido ix) edta^{4-} x) NO^+ xi) NO_2^- ligand xii) Cyanato ligand
 xiii) isothiocyanato ligand xiv) Cyano ligand xv) Carbonato ligand

Q3. What is chelate complex? Draw the chelated complex structure of the following ligands with metallic atom (M). And how much chelated rings are formed with following each ligand? (07 marks)

- a) M-Triethylene tetramine b) M-diethylene triamine c) M-triaminotriethylenamine d) M-tetraethylene
 pentamine e) M-ethylene diamine triacetate ion f) Nickel with dimethylglyoximate

Q4. Fill in the blanks for each of the following: (02 marks)

- i. Isomersim exhibited by the complex $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{SO}_4$ is -----
 ii. The pair of complexes $[\text{Cr}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}]\text{Cl}_2$ and $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}\cdot\text{H}_2\text{O}$, illustrates-----
 iii. The number of isomers possible for a square planar complex $[\text{Mabcd}]$, is-----
 iv. The kind of isomerism illustrated by complexes containing diaminopropane as one of the ligand is-----

Q5. Give the best answer of the following: (3.5 marks)

- i. Sketch the geometric isomers for the square planar complex, $[\text{Pt}(\text{NH}_3)_2\text{ClBr}]$
 ii. How many isomers are possible for the neutral complex $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$? Sketch them.
 iii. Sketch the possible geometric isomers for $[\text{Cr}(\text{NH}_3)_2\text{Cl}_4]^-$
 iv. Match the following:

$[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$	a) Hydrate isomers
$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]\text{Br}_2$ and $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$	b) Coordination isomers
$[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2\cdot\text{H}_2\text{O}$ and $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}\cdot 2\text{H}_2\text{O}$	c) Linkage isomers
$[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$ and $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$	d) Ionization isomers

Inorganic :: Sir Sattar Chah ^{#CR} Agha

ASSIGNMENT BS-III, MSc (prev) First Semester 2019

Q. Explain the following with appropriate reasons.

1. $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ exchanges two of its water molecules more easily than the other four.
2. $[\text{Cu}(\text{NH}_3)_6]^{2+}$ is deep blue coloured whereas $[\text{CuCl}_4]^{2-}$ is yellow coloured even though both contains Cu^{2+} ions. —
3. $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is a stronger oxidising agent than $[\text{Fe}(\text{CN})_6]^{3-}$.
4. 4d and 5d series transition elements form low spin complexes.
5. A number of tetrahedral $\text{Co}(\text{II})$ complexes, unlike tetrahedral $\text{Ni}(\text{II})$ complexes, are stable. —
6. Aqueous solution of $\text{Ni}(\text{II})$ is green colored whereas that of $\text{Zn}(\text{II})$ is colorless.
7. CO or CN^- is a blood poison.
8. Color of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is blue.
9. Cr^{2+} does not form a regular octahedral complex with a weak field ligand whereas Mn^{2+} forms a regular octahedral complex with that ligand.
10. Cr^{3+} and Co^{3+} form generally inert octahedral complexes.
11. Cu^{2+} does not form a regular octahedral complex whereas Ni^{2+} does.
12. Cu^{2+} ions are colored and paramagnetic while Zn^{2+} ions are colorless and diamagnetic.
13. F^- , inspite of being smallest in size, does not give the strongest field of all ligands.
14. H_2O ligand gives higher crystal field splitting than OH^- ligand.
15. High spin octahedral complexes of Mn^{2+} ions are colorless.
16. In tetrahedral case t_{2g} orbitals are higher in energy while in octahedral case these are lower in energy than e_g orbitals.
17. In the crystal of CuF_2 , all the $\text{Cu}-\text{F}$ bond distances are not equal.
18. Jahn-Teller distortion may lead either to "Z-out case" or to "Z-in case". Generally "Z-out case" is much more common than "Z-in case".
19. $\text{K}_2[\text{NiF}_6]$ is diamagnetic while $\text{K}_3[\text{CrF}_6]$ is paramagnetic.
20. MnBr_4^{2-} is paramagnetic with a moment of 5.92 BM.
21. Octahedral complexes with three and eight d -electrons on the central metal atom or ion are particularly stable.
22. Octahedral Field splitting energy is always higher than tetrahedral field splitting energy.
23. Pt^{4+} ion forms octahedral where as Pt^{2+} forms square planar complexes.
24. Splitting Energy of d orbitals by CN^- ligand is greater than that of F^- ligand.
25. Square planar geometry is especially common for d^8 configuration.
26. Tetrahedral complexes are high spin.
27. Though the normal oxidation state of Ni is +2, $[\text{Ni}(\text{CO})_4]$ with $\text{Ni}(0)$ is stable.

The End

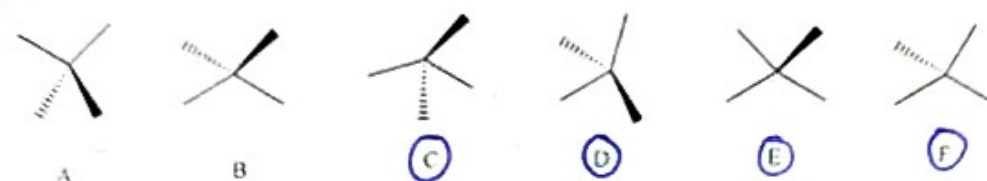
Name of Student Shanzah Roll No. 2K20/CHE/133 BS/MSc (Prev.)

NOTE: ATTEMPT ALL QUESTIONS

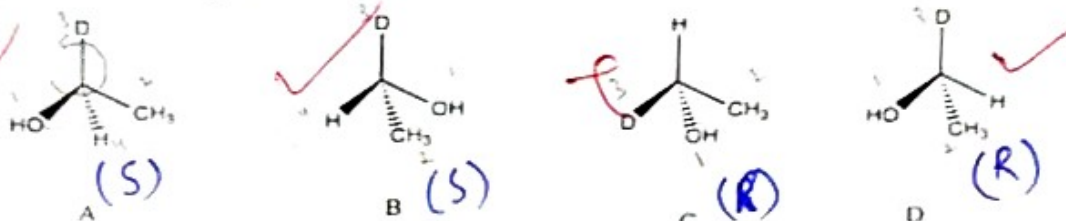
Q.1: Which of the following is/are meso compound(s)?

- A) 2,4-dichloropentane B) 1,3-dimethylcyclopentane
C) 2,3-dichloropentane D) I and II

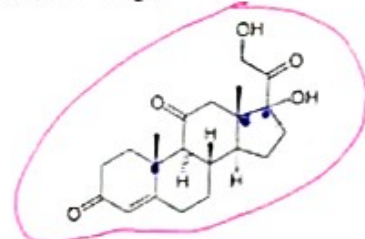
Q.2: Encircle incorrect 3D structure(s) in the followings;



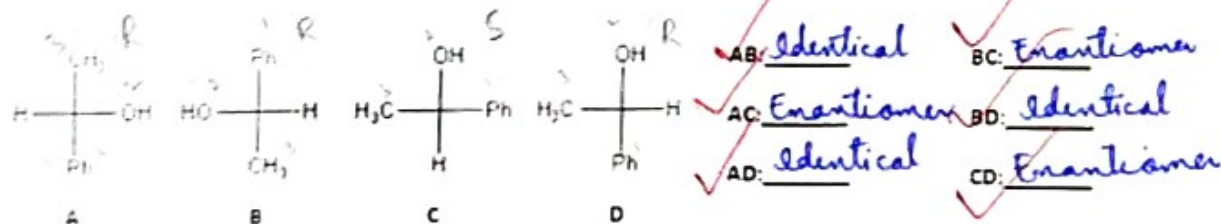
Q.3: Predict R/S configuration of chiral center in each of the following compounds



Q.4: Identify the chiral centers in following structure



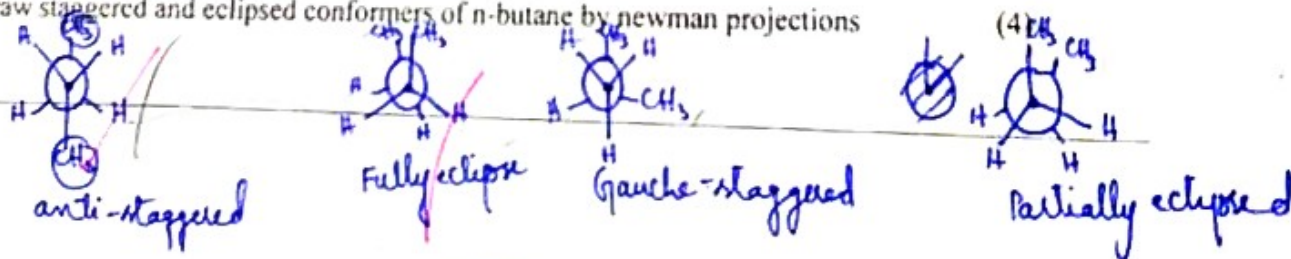
Q.5: Predict the stereochemical relations in the following pairs;



Q.6: Which of the following is/are cis and trans forms of dimethylcyclohexane?

- A) 1,2-Diaxial trans C) 1,3-Diaxial cis E) 1,4-Diaxial trans
B) 1,2-Diequatorial trans D) 1,3-Diequatorial cis F) 1,4-Diequatorial trans

Q.7: Draw staggered and eclipsed conformers of n-butane by newman projections



27.5
30

Good

Highest
score

11.5

1.8

1.6

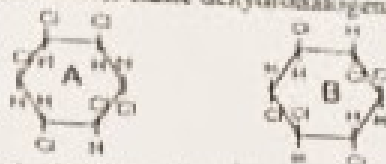
1.4

Q. NO. 3: Explain why. Attempt any FIVE of the followings. [20 Marks]

- Hydrolysis of CH_3Cl is catalyzed by addition of NaI .
- Hydrolysis of tertiary butyl chloride is retarded by addition of NaCl .
- Optical activity of 2-iodooctane slowly falls to zero when allowed to react with KI in acetone.
- Following compounds are notoriously slow to both $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reaction



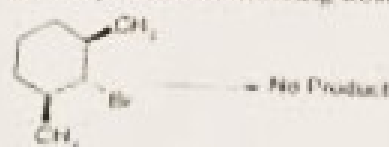
- Which of the following would give faster basic dehydrohalogenation and why.



- The rate of formation of ter-butyl ethyl ether from ter-butyl bromide does not increase by changing the nucleophile from $\text{C}_2\text{H}_5\text{OH}$ and $\text{C}_2\text{H}_5\text{ONa}$ which is better nucleophile.
- In substitution reaction, isopropyl chloride reacts faster with diethylamine than with ammonia, whereas diphenylmethyl chloride reacts essentially at the same rate with both nucleophile.
- Dehydration of alcohol becomes facile as we go from ethanol, 2-propanol, 2-methyl-2-propanol.
- Which of the followings is best way to prepare isopropyl methyl ether.



- Explain the following observations and write the products wherever required



- What is correlation between sign of optical activity and R/S configuration.
- All optically active compounds must be asymmetric but all asymmetric molecules need not be optically active. Give examples.

Dr. M. A. Kazi Institute of Chemistry, University Of Sindh, Jamshoro
BSc III & MSc. (Prev) Inorganic Chemistry (A-group)
Final Test Date: (24/05/2022) Total Marks: 50 Time: 40 minutes

Note: Select the most appropriate option.

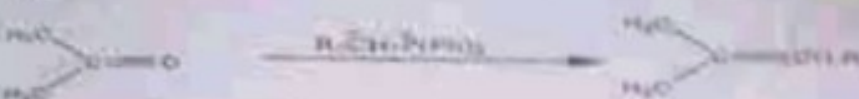
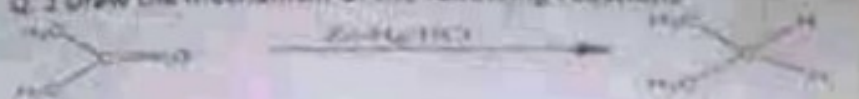
- The CFSE for a high-spin d^5 octahedral complex is: (a) $-0.6 \Delta_o$ (b) $1.2 \Delta_o$ (c) $1.8 \Delta_o$ (d) $-0.4 \Delta_o$
- Which metal complex ion is expected to be subject to a Jahn-Teller distortion? (a) $[\text{Cr}(\text{OH})_6]^{3+}$ (b) $[\text{Cr}(\text{NH}_3)_6]^{2+}$ (c) $[\text{Cr}(\text{CN})_6]^{3-}$ (d) $[\text{Cr}(\text{bpy})_3]^{2+}$
- What is the spin-only magnetic moment for high-spin $[\text{Fe}(\text{ox})_3]^{3-}$? (a) 4.90 Bm (b) 2.83 B (c) 5.92 Bm (d) 3.87 Bm
- In the complex ion, $[\text{Cr}(\text{C}_2\text{O}_4)_2(\text{H}_2\text{O})_2]^-$ oxidation state of chromium is (a) $+1$ (b) $+2$ (c) $+3$ (d) $+4$
- The complex with spin only magnetic moment of 4.9 B.M. is (a) $[\text{Fe}(\text{CN})_6]^{3-}$ (b) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (c) $[\text{Fe}(\text{CN})_6]^{4-}$ (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
- The value of spin only magnetic moment for one of the following configurations is 2.84 B.M. . The correct one is (a) d^4 (in strong field ligand) (b) d^4 (in weak field ligand) (c) d^3 (in strong as well as in weak field ligand) (d) d^5 (in strong field ligand)
- How many unpaired electrons are there in a strong field iron (II) octahedral complex? (a) 0 (b) 1 (c) 2 (d) 4
- The ion which is expected to have Jahn-Teller distortion in an octahedral field is (a) Co^{3+} (strong field) (b) Ni^{2+} (c) Mn^{2+} (weak field) (d) Co^{2+} (low spin)
- Which of the following correctly places the ligands in their order in the spectrochemical series? (a) $\text{Br}^- < \text{Cl}^- < \text{NH}_3 < \text{H}_2\text{O}$ (b) $\text{I}^- < \text{Br}^- < \text{H}_2\text{O} < [\text{OH}]^-$ (c) $\text{F}^- < \text{Cl}^- < \text{H}_2\text{O} < \text{NH}_3$ (d) $\text{I}^- < \text{Cl}^- < \text{H}_2\text{O} < \text{en}$
- The CFSE for a high-spin d^6 octahedral complex is (a) $-0.6 \Delta_o$ (b) $1.2 \Delta_o$ (c) $1.8 \Delta_o$ (d) $-0.4 \Delta_o$
- The CFSE for a high-spin octahedral complex of a d^7 ion is (a) $2.4 \Delta_o$ (b) $-1.8 \Delta_o$ (c) $1.2 \Delta_o$ (d) $-0.6 \Delta_o$
- Which statement is CORRECT? (a) A low spin complex results when Δ_o exceeds the energy required to pair electrons, P. (b) A high spin complex is never paramagnetic. (c) Ligands like NO_2^- and CN^- usually produce high spin complex ions. (d) A low spin complex ion is always diamagnetic.
- Which one of the following statements is FALSE? (Crystal Field Theory) (a) In an octahedral crystal field, the d electrons on a metal ion occupy the eg set of orbitals before they occupy the t2g set of orbitals. (b) Diamagnetic metal ions cannot have an odd number of electrons. (c) Low spin complexes can be paramagnetic. (d) In high spin octahedral complexes, CFSE is less than the electron pairing energy, and is relatively very small. (e) Low spin complexes contain strong field ligands.
- Which is the strongest field ligand? (a) NH_3 (b) F^- (c) H_2O (d) CN^-
- Which of the following is a π -donor ligand? (a) Cl^- (b) NH_3 (c) CO (d) PF_3
- Coordination number of metal ion represents (a) no. of d-orbital represents (b) number of sigma bond formed by metal ion (c) no. of pi bonds formed by metal ion (d) none of these

Total Marks: 70, allowed 2.0 hr. Dated: 11.12.2018

Note: Attempt all Questions.

Q. 1 Draw the mechanism of the following reactions

(20x2=20)



Q. 2 Suggest the reason to the following facts

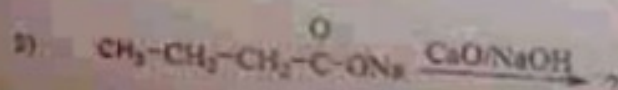
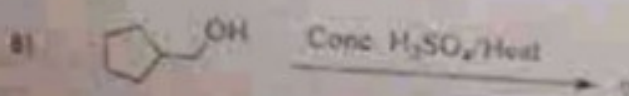
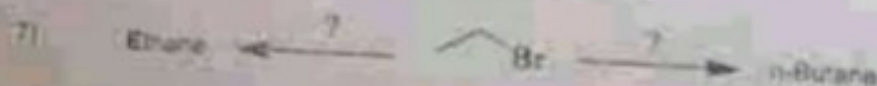
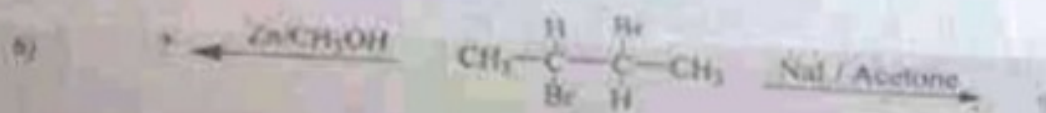
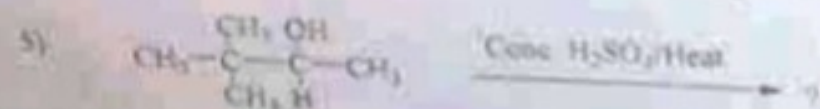
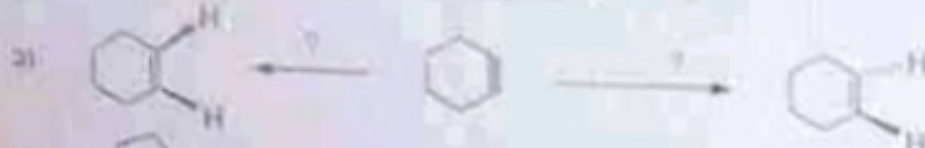
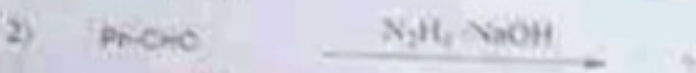
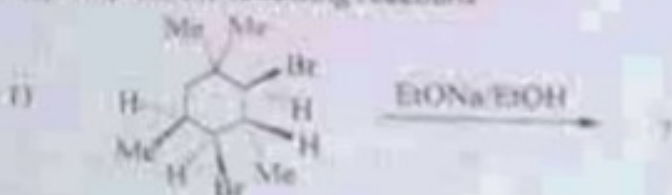
(10x2=20)

a) Phenol is more acidic than alcohol

b) Aniline is less basic than ethylamine

Q. 3 Complete the following reactions

(20)



Q. 4 Define the following terms

a) Conjugation

b) Hyperconjugation

c) Localized bond

d) Delocalized bond

e) Inductive effect

(5x2=10)

DR. M.A. KAZI INSTITUTE OF CHEMISTRY, UNIVERSITY OF SINDH, JAMSHORO

Second Semester

BS II (Morning & Evening)

Date: 12-12-2015

Failures/Impro

Computing in Chemistry

COMP: 401

Time allowed: 2.5 h

Max. Marks: 100

Note: Attempt any FIVE questions from the following:

- Q. 1: Delineate the computational chemistry. Draw the cycle and explain in detail the steps utilized by a chemist in confronting and solving the problems.
- Q. 2: Describe the parts of computer system?
- Q. 3: Explain: Speed, Accuracy, Record, Secrecy, Diligency, Capability and Flexibility of the computer system?
- Q. 4: Define with examples the system, data, information and interpretation?
- Q. 5: Write down the possible short keys and their functions usually frequently used by a chemist?
- Q. 6: What is software? Enlist the softwares used in the field of chemistry? And also state classification according to their applications?
- Q. 7: Distinguish between Ms. Office and advanced softwares.

(This end & wishing you the best of luck)

IAZI INSTITUTE OF CHEMISTRY, UNIVERSITY OF SINDH, JAMSHORO

Semester

BS II (Morning & Evening)

Date: 16-12-2014

Star

Computing in Chemistry

COMP: 401

Time allowed: 2 & 1/2 h

Max. Marks: 90

Note: Attempt any six questions from the followings:

Q. 1: Delineate computational chemistry? Discuss the problems faced by the chemist? And also write down the steps through which a chemist may able to solve them.

Q. 2: Depict the parts of computer system?

Q. 3: Explain: Speed, Accuracy, Record, Secrecy, Diligency, Capability and Flexibility of the computer system?

Q. 4: Define with examples the system, data, information and interpretation?

Q. 5: Inscribe the possible short keys and their functions usually frequently used by a chemist?

Q. 6: What is software? Enlist the softwares used in the field of chemistry? And also state classification according to their applications?

Q. 7: Discriminate between: (i) computer and calculator (ii) Ms. Office and advanced softwares.

MAZAH INSTITUTE OF CHEMISTRY, UNIVERSITY OF SINDH, JAMSHORO

BS II (Morning & Evening)

Date: 13-12-2013

Computing in Chemistry

COMP. 401

Semester

Year

Time allowed: 2 h

Max. Marks: 50

Note: Attempt ~~any four~~ ^{all} questions from the following:

- Q. 1: What do you mean by computing in chemistry? Discuss the problems faced by the chemist? And also write down the steps through which a chemist may be able to solve them.
- Q. 2: Illustrate the parts of computer system?
- Q. 3: What are the advantages and disadvantages of the computer?
- Q. 4: Define with examples the system, data, information and interpretation?
- Q. 5: Write the possible short keys that are usually frequently used by a chemist?
- Q. 6: Differentiate between: (I) computer and calculator (II) Ms. Office and advanced software.

(The End)

DR. M. A. KAZI INSTITUTE OF CHEMISTRY, UNIVERSITY OF SINDH, JAMSHORO

2nd Semester *BS II (Morning)* Regular & I/F

Computing in Chemistry (COMP 401)

Date: 14-12-2019

Time allowed: 1 h

Max. Marks: 90

Q. 1: Explain the below contents in detail & also discuss the inter-connectivity among them along with the basis of Computational Chemistry.

1. *Identify & define the problems.*
2. *Design the experiments.*
3. *Conduct the experiments & gather the data.*
4. *Analyze the data.*
5. *Propose a solution to the problems.*

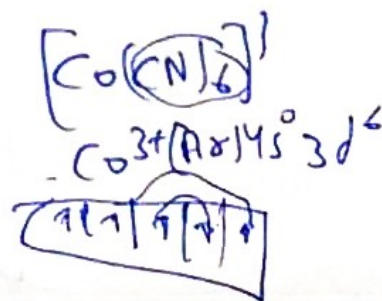
Q. 2: Define with examples the system, data, information and interpretation?

Q. 3: What is software? Enlist the softwares used in the field of chemistry? And also state classification according to their applications?

(The end & wishing you the best of luck)

17. Given $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ (high spin). Calculate ΔCFSE and give configuration in term of t_2g and eg
 (a) $2.6 \Delta_0, t_2g^1 eg$ (b) $1.2 \Delta_0, t_2g^6 eg^2$ (c) $1.2 \Delta_0, t_2g^6 eg^2$ (d) $2.6 \Delta_{sp}, t_2g^5 eg^2$
18. The correct increasing order of splitting power of ligands according to spectrochemical series is
 (a) $\text{Cl}^- < \text{OH}^- < \text{CN}^-$ (b) $\text{Cl}^- < \text{CN}^- < \text{OH}^-$
 (c) $\text{OH}^- < \text{Cl}^- < \text{CN}^-$ (d) $\text{OH}^- < \text{CN}^- < \text{Cl}^-$
19. Which statement regarding $[\text{Cr}(\text{NH}_3)_6]^{3+}$ is incorrect?
 (a) It has octahedral geometry (b) It has d^2sp^3 hybridization
 (c) It is diamagnetic (d) It is a low spin complex
20. What is the number of unpaired electrons in $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ if it is known to be a high spin complex?
 (a) 0 (b) 2 (c) 3 (d) 4
21. The crystal field theory considers the metal-ligand bond to be a _____ bond.
 (a) covalent (b) ionic (c) polar (d) dative
22. Which of the following orbitals does not belong to t_2g orbitals?
 (a) d_{xy} (b) d_{yz} (c) d_{xz} (d) d_{z^2}
23. Identify the correct relation between Δ_0 and Δ_t , where Δ_0 denotes crystal field splitting in octahedral complexes and Δ_t denotes crystal field splitting in tetrahedral complexes.
 (a) $\Delta_0 < \Delta_t$ (b) $\Delta_0 > \Delta_t$ (c) $\Delta_0 = \Delta_t$ (d) $\Delta_0 \geq \Delta_t$
24. Which of the following ligands is most likely to form high spin complexes in octahedral fields?
 (a) Cl^- (b) OH^- (c) $\text{C}_2\text{O}_4^{2-}$ (d) CN^-
25. What will be the electronic configuration of d in terms of t_2g and eg in an octahedral field when $\Delta_0 < P$, where P is the energy required for pairing of electrons in a single orbital?
 (a) $t_2g^5 eg^0$ (b) $t_2g^2 eg^3$ (c) $t_2g^3 eg^2$ (d) $t_2g^0 eg^5$
26. As the crystal field splitting energy in octahedral field increases, the wavelength of light absorbed
 (a) Increases (b) Decreases (c) Remains the same (d) May increase or decrease
27. Identify the correct naming for $\text{K}_2[\text{PdCl}_4]$
 (a) Potassium tetrachlorinepalladium (II)
 (b) Potassium tetrachlorinepalladate (II)
 (c) Potassium tetrachloridopalladium (II)
 (d) Potassium tetrachloridopalladate (II)
28. Identify the correct formula for potassium tetracyanonickelate(II)
 (a) $\text{K}[\text{Ni}(\text{CN})_4]$ (b) $\text{K}[\text{Ni}(\text{CN})_4]_2$ (c) $\text{K}_2[\text{Ni}(\text{CN})_4]$ (d) $\text{K}_2[\text{Ni}(\text{CN})_4]_2$
29. The sum of coordination number and oxidation number of the metal M in the complex $[\text{M}(\text{en})_2(\text{C}_2\text{O}_4)]\text{Cl}$ (where en is ethylenediamine) is
 (a) 9 (b) 6 (c) 7 (d) 8
30. A chelating agent has two or more than two donor atoms to bind to a single metal ion. Which of the following is not a chelating agent?
 (a) Thiosulphato (b) Oxalato (c) Glycinato (d) Ethane-1,2-diamine
31. In the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4)]\text{NO}_2$ (where en is ethylenediamine) _____ are the coordination number and the oxidation state of the element 'E' respectively.
 (a) 6 and 2 (b) 2 and 2 (c) 4 and 3 (d) 6 and 3
32. How many ions are produced from $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ in solution.
 (a) 3 (b) 4 (c) 2 (d) 5
33. Which of the following cyano complexes would exhibit the lowest value of paramagnetic character?
 (a) $[\text{Cr}(\text{CN})_6]^{3-}$ (b) $[\text{Mn}(\text{CN})_6]^{3-}$ (c) $[\text{Fe}(\text{CN})_6]^{3-}$ (d) $[\text{Co}(\text{CN})_6]^{3-}$

Best of Luck



SECTION-B: (BRIEF QUESTIONS)

Note: Attempt all the questions from this section.

Q1. Name the following complexes (5 marks)

- i. $[\text{Co}(\text{NH}_3)_5\text{CO}_3]\text{Cl}$ ii. $[\text{Pt}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_2\text{Br}_2]$ iii. $[\text{Cr}(\text{NCS})_4(\text{NH}_3)_2]$ iv. $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$
v. $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$ vi. $\text{K}_3[\text{In}(\text{C}_2\text{O}_4)_3]$ vii. $[(\text{NH}_3)_4\text{Co}-(\text{NH}_2)_2\text{O}_2-\text{Co}(\text{NH}_3)_4]^{2+}$
viii. $[(\text{NH}_3)_3\text{Co}-\text{NH}_2-\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})]\text{Cl}_3$ ix. $[(\text{C}_2\text{O}_4)_2\text{Cr}-(\text{OH})_2-\text{Cr}(\text{C}_2\text{O}_4)_2]^{4-}$ x. $[\text{Cr}(\text{en})_3][\text{Ni}(\text{CN})_5]$

Q2. Write the formula/structure of the following (5 marks)

- i. tetraamminesulphatocobalt(III) nitrate ii. sodium tetrachlorozincate(II) iii. sodium amminebromochloronitroplatinate(II) iv. dichlorobis(ethylenediamine)nitritocobalt(III) chloride v. octacarbonyl-μ-dicarbonyldiron(0)
vi. bis(acetylacetonato)oxovanadium(IV) vii. sodium oxotetrafluorochromate(IV)
viii. potassium tetrafluoroargentate(I) ix. tetrabromo-μ-dibromodibromodipalladium(II) ion x. hexaaquairon(II) sulphate

Q3. What is CFSE? Calculate the CFSE in terms of Δ_0 for octahedral complexes for d^4 to d^7 electron configuration in case of high-spin and low-spin complexes. (2.5 marks)

Q4. Draw the crystal field splitting energy diagram of octahedral, tetragonal and square planar complex (2.5 marks)

Q5. Describe the Platinum and Cobalt complexes by Werner's in the tabulated format.

Hint: Old formula, Werner's Formula, Dissociation, number of particles, and number of electrical charge (2.5 marks)

Q6. On the basis of VBT, predict the oxidation state of the central metal atom/ion, hybridization, geometry and magnetic behavior of the following complexes. (2.5 marks)

- a) $[\text{Zn}(\text{CN})_4]^{2-}$ b) $[\text{Ni}(\text{NH}_3)_4]^{2+}$ c) $[\text{CoCl}_4]^{2-}$ d) $[\text{PtCl}_4]^{2-}$ e) $[\text{Fe}(\text{CN})_6]^{3-}$

Q7. Predict the outer-orbital/inner-orbital octahedral complexes by VBT (2.5 marks)

- a) $[\text{Fe}(\text{NH}_3)_6]^{2+}$ b) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ c) $[\text{MnF}_6]^{2-}$ d) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ e) $[\text{Fe}(\text{CN})_6]^{4-}$

Q8. Write the oxidation state of the central metal in each of the following. (2.5 marks)

- a) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]^+$ b) $\text{K}[\text{CuBr}_2]$ c) $[\text{Co}(\text{py})_2\text{SO}_4\text{Cl}]$ d) $[\text{Cr}(\text{ox})_3]^{3-}$ e) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2\text{H}_2\text{O}$

Q9. What is the EAN? Calculate EAN for the central metal atom in the following complexes and also compare the stability of the complexes (2.5 marks)

- a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ b) $\text{Cu}(\text{CN})_4]^{2-}$ c) $[\text{Fe}(\text{CN})_6]^{4-}$ d) $[\text{Fe}(\text{CN})_6]^{3-}$ e) $[\text{Ag}(\text{NH}_3)_2]^+$

Q10. Define are flexidentate and ambidentate ligands give an example with structures? Write denticity of the ligands in the following complexes. (3.5 marks)

- a) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ b) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ c) $[\text{Ni}(\text{dmg})_2]$ d) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Cl}_2$ e) $\text{Na}_2[\text{Ni}(\text{edta})]$

Q11. Arrange the following in the increasing order of molar conductivity by Werner's. (01 mark)

- a) $\text{Na}[\text{Cr}(\text{NH}_3)_2\text{Cl}_4]^{2-}$ b) $\text{Cr}(\text{NH}_3)_3\text{Cl}_3$ c) $\text{Na}_2[\text{CrCl}_4]$

Q12. A complex of the composition $\text{PtCl}_4(\text{NH}_3)_2$ has a molar conductance of 404 siemens. Based on this data, guess its proper formula. (01 mark)

Q13. The substance $\text{CoBr}_3 \cdot 4\text{NH}_3 \cdot 2\text{H}_2\text{O}$ behaves like a $(3+1-)$ electrolyte in solution; write its correct formula (01 mark)

Q14. Explain why Cu^{2+} does not form a regular octahedral complex whereas Ni^{2+} does. (01 mark)

Q15. Since NH_3 is in the middle of the spectrochemical series, some of its complexes are high-spin while others are low-spin. Illustrate. (01 mark)

Q16. Calculate the CFSE in terms of Δ_0 for a d^8 system in octahedral and tetrahedral complexes. Which is more stable? (01 mark)

Q17. Draw the splitting diagram of d^8 electrons of Cu^{2+} ion in tetragonally elongated and compressed distorted octahedral complexes. Also calculate the Jahn-Teller stabilization energy (02 marks)

Q18. Draw the diagram for the splitting of d-orbitals of d^8 ion in its low-spin and high-spin tetragonally elongated distorted octahedral complexes. Which complex show strong distortion? (01 mark)

Dr. M. A. KAZI INSTITUTE OF CHEMISTRY

INORGANIC CHEMISTRY, 1st SEMESTER, EXAMINATION 2022
BS (P-4), GROUP- 5, REGULAR, MORNING BATCH

SUBJECT: INORGANIC CHEMISTRY THEORY

Date: 25 - 5 - 2022

Time Allowed: 2hrs

COURSE # CHEM-500

Max. Marks: 80

Q1. Explain the following statements with suitable answers (20 Marks)

- Amongst the two, $[\text{Co}(\text{NO}_2)_6]^{3-}$ ion and $[\text{Co}(\text{NO}_2)_6]^{3+}$ ion, which would absorb light of longer wavelength in the visible region?
- An aqueous solution of zinc nitrate contains the ion, $[\text{Zn}(\text{H}_2\text{O}_6)]^{2+}$ and is colorless. What do you infer about the absorption spectrum of the $[\text{Zn}(\text{H}_2\text{O}_6)]^{2+}$ ion?
- Should the ion, $[\text{Co}(\text{NO}_2)_6]^{3-}$ be easy or difficult to oxidize to $[\text{Co}(\text{NO}_2)_6]^{4-}$. Explain your answer.
- Jahn-Teller distortion may lead either to 2-out case or to 2-in case. Generally, 2-out case is much more common than 2-in case. Why?
- Octahedral coordination complexes with three and eight d-electrons on the central metal atom/ion are particularly stable. Under what condition, would you expect a complex with six d-electrons on the central metal atom/ion to be particularly stable. Explain why?
- Though the normal oxidation state of Ni^{2+} , $[\text{Ni}(\text{CO})_4]$ with NiCl_2 is stable. Explain why?
- $[\text{Fe}(\text{CN})_6]^{3-}$ has only one unpaired electron whereas $[\text{Fe}(\text{CN})_6]^{4-}$ has five unpaired electrons. Explain why?
- $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ is diamagnetic whereas $[\text{Co}(\text{F})_6]^{3-}$ is paramagnetic. Explain why?
- F^- instead of being strongest in size, does not give the strongest field of all ligands. Explain.
- Tetrahedral $\text{Cu}(\text{II})$ complexes are stable while $\text{Ni}(\text{II})$ prefer to form square planar complexes. Explain why?

Q2 (a). Write down Werner's formula for the platinum complex, $\text{PtCl}_4(\text{NH}_3)_2$ (05 marks)

Where $n = 0, 2, 3, 4, 5$ and 6

How do they dissociate in water?

Q2 (b). Write down Werner's formula (05 Marks)

i) $\text{PtCl}_2(\text{NH}_3)_2 \cdot \text{KCl}$ ii) $\text{PtCl}_2 \cdot 2\text{KCl}$

How do they dissociate in water?

Q2 (c). Write down Werner's data and structure on complexes of ammonia with CrCl_3 (05 Marks)

Q3 (a). Draw the crystal field splitting of d-orbitals in regular Octahedral, elongated/compressed tetragonal and square planar complexes? And write the order of increasing energy. (05 marks)

Q3 (b). For an octahedral Mn^{2+} complex, $\Delta_o = 21,150 \text{ cm}^{-1}$ and electron pairing energy, $E_p = 27750 \text{ cm}^{-1}$, will it be a high spin or low spin complex? Calculate its CFSE. Find out the spin only magnetic moment of that complex ion. (05 marks)

Q3 (c). Prove the splitting energy of square planar is greater than octahedral ($\Delta_{sp} = 1.70 \Delta_o$). (05 marks)

Q4 (a). Explain. $\text{Au}(\text{I})$ ion is less stable even though the $\text{Cu}(\text{I})$ and $\text{Ag}(\text{I})$ ions are comparatively more stable. Why do the d^9 system usually not under complete distortion to square planar geometry? (05 marks)

Q4 (b). High spin $\text{Fe}(\text{II})$ complexes like $[\text{FeCl}_4]^{2-}$ are tetrahedral but $[\text{PdCl}_4]^{2-}$ is square planar. Explain why? (2.5 marks)

Q4 (c). On the basis of VBT explain the geometry and magnetic properties of the following complexes. (2.5 marks)

i) $[\text{CuCl}_4]^{2-}$ ii) $[\text{Ni}(\text{CO})_4]$ iii) $[\text{Fe}(\text{CO})_5]$ iv) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ v) $[\text{Co}(\text{NH}_3)_6]^{3+}$

Note: Attempt all questions. Give Brief and to the point answers.

Q. 1: Give brief answers to any EIGHT of the followings. [3x8 = 24 marks]

1. Briefly define following terms.

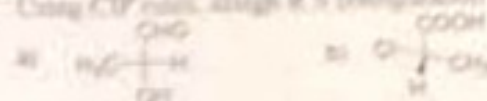
[a] Stereochemistry [b] Stereoisomers [c] Enantiomers

2. What is relationship in followings

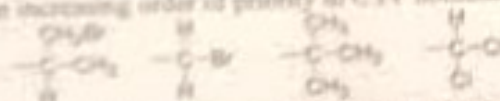
[a] Asymmetry [b] Chirality [c] Optical activity

3. Draw the Fischer projection for Tartaric acid and assign stereoisomeric relationship to them

4. Using CIP rules, assign R/S configuration to followings



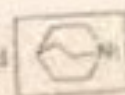
5. Arrange the following in increasing order of priority in CIP nomenclature.



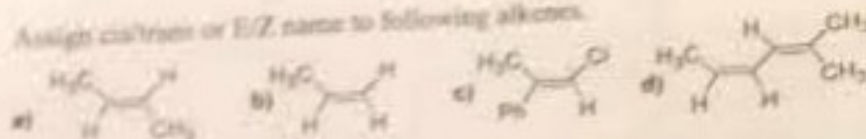
6. What are the crucial requirements for elimination reactions to occur.

7. Which nucleophile is stronger in each pair.

(a) RO^- and ROH (b) HO^- and HS^- (c) $(\text{C}_2\text{H}_5)_2\text{N}$ and



8. Assign cis/trans or E/Z name to following alkenes.



10. Differentiate the following pairs

(a) Lewis Base & Nucleophile (b) Electrophile and Nucleophile

13. How optical activity can be used to distinguish enantiomers, diastereomers and meso compounds

Q. 2: Give brief answers to any FIVE of followings: [3x5 = 15 marks]

i. Highlight main points of transition state

ii. Define reactive intermediates give examples

iii. Point out main rules of Fischer projection

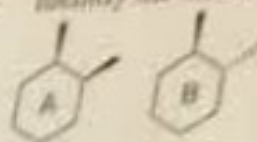
iv. Draw and name the conformers of ethane as Newman projection and give them names

v. What is condition for molecules to exhibit geometric isomerism?

vi. List out the major types of organic reactions with examples

vii. What are two types of polar solvents.

viii. Identify the meso compound in following pair.



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Final Test: BS-III, 1st Semester 2022

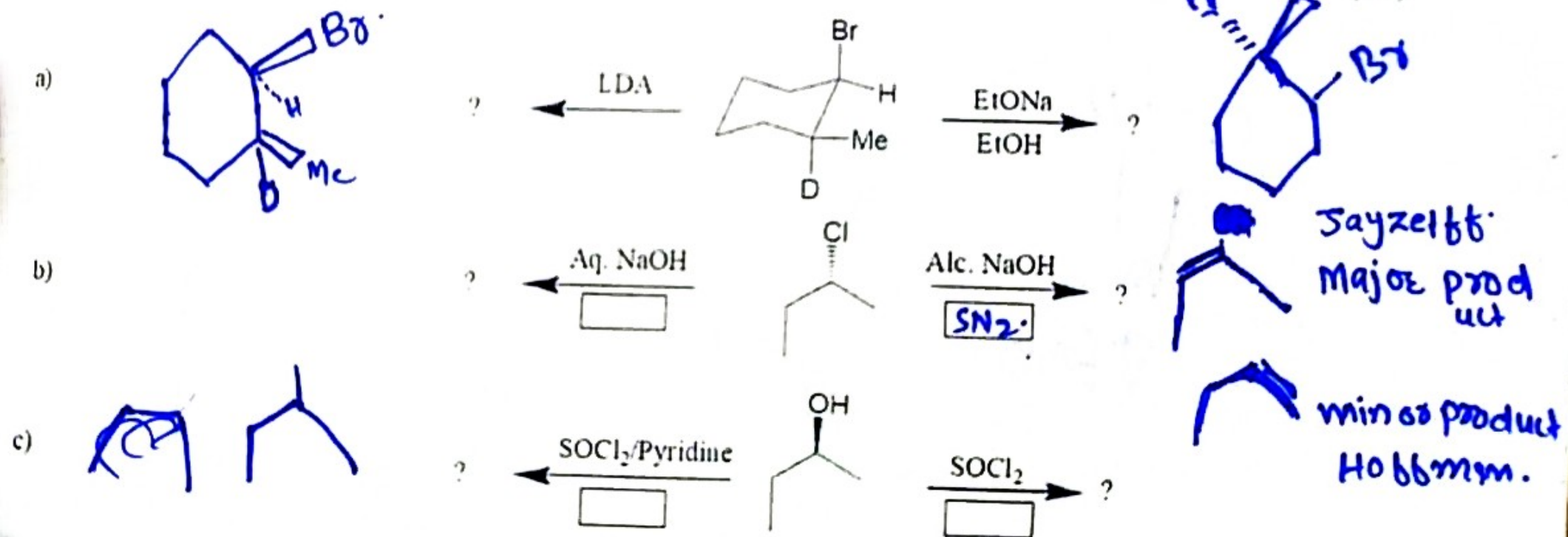
Maxim Mark: Final test: 50, Mid-Term Test: 30, Assignment: 10 & Attendance: 10

Time allowed 60 minutes, Dated: 27-05-2022

Name of Student _____ Roll No. _____ BS/MSc (Prev.)

NOTE: ATTEMPT ALL QUESTIONS

Q. 1 Complete the Following reactions



Q. 2a Identify the stereochemical relations between following



Answer:

Diastomers

(5)